Western Ghats Deforestation & Land Cover Change Detection

Region: **Sakleshpur |** Years: 2022 → 2026

Generated: 09 June 2026 09:23

Alert Level: Low Risk

Forest Cover Summary

Metric	Value
Forest Cover (Year 1)	45.76%
Forest Cover (Year 2)	52.66%
Forest Loss	0.00%
Forest Gain	6.90%
Net Forest Change	+6.90%

Land Cover Percentages

Class	Year 2022 %	Year 2026 %	Change %
Forest	45.76%	52.66%	+6.90%
Agriculture	0.01%	0.01%	+0.00%
Urban	43.82%	40.25%	-3.57%
Water	10.23%	6.22%	-4.02%
Barren	0.18%	0.87%	+0.68%

Key Land Cover Transitions

Transition	Area (%)
Forest → Urban	3.752%
Forest → Agriculture	0.000%
Forest → Barren	0.031%
Agriculture → Urban	0.000%

Recommendations

• Urban expansion detected near forest boundary.

Academic Analysis & Ecological Justification

1. Reasoning for Land Cover Shifts

A positive trend with a 6.90% increase in forest cover suggests effective local conservation efforts and natural regeneration, aligning with recent afforestation drives under the Green India Mission. A concerning reduction of 4.02% in water bodies correlates with reduced groundwater recharge due to catchment degradation, exacerbating hydrological vulnerability in the region as noted by the Central Water Commission. The 0.68% increase in barren land highlights soil erosion and localized resource extraction (such as quarrying), which strip the land of its natural ecological resilience.

2. Steps for Ecological Balance

• **Enforcement of Eco-Sensitive Zones (ESZs):** Strict implementation of zoning laws as recommended by the Kasturirangan Committee to restrict hazardous industrial and mining activities in vulnerable tracts.

• **Agroforestry Integration:** Transitioning from monoculture plantations to biodiversity-friendly agroforestry systems in buffer areas to restore soil fertility and create wildlife corridors.

• **Hydrological Restoration:** Rejuvenating degraded catchments through watershed management and afforestation to secure the water table and sustain regional hydrology.

3. Conclusive Points

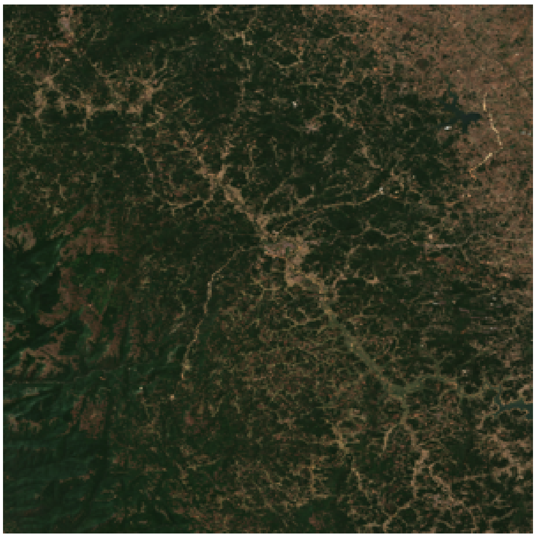
• **Habitat Fragmentation & Edge Effects:** The observed transitions in Sakleshpur reveal a concerning pattern. The conversion of natural habitats to anthropogenic land uses isolates endemic species and diminishes overall biodiversity capacity.

• **Ecological Vulnerability:** The land-cover dynamics underscore a precarious balance. Without proactive management, even minor anthropogenic perturbations could trigger irreversible ecological decline in this global biodiversity hotspot.

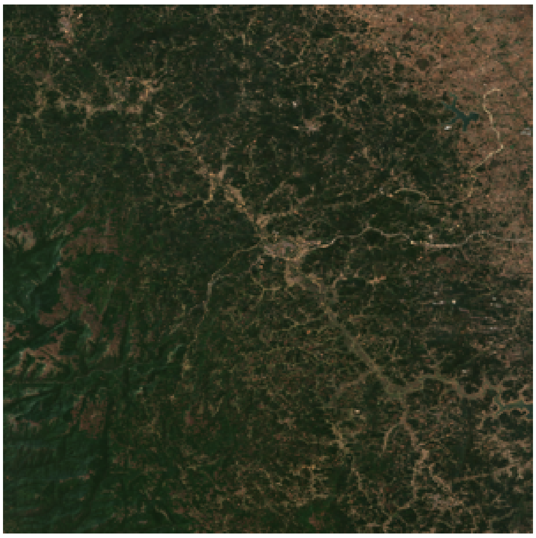
• **Imperative for Data-Driven Policy:** Real-time, satellite-derived insights emphasize the critical need for executing ground-level environmental regulations promptly. Continued unmonitored land-cover alterations will irreversibly compromise the long-term ecological and economic stability of peninsular India.

Visual Maps

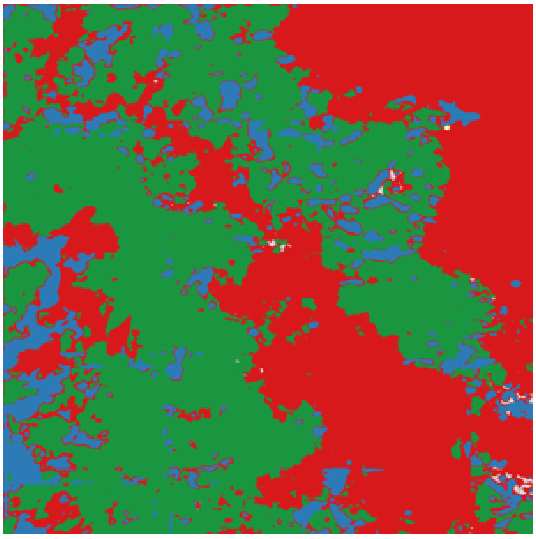
Satellite Image — Year 2022



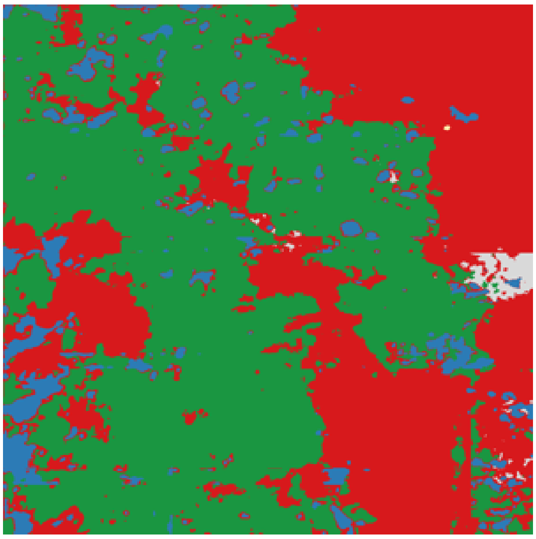
Satellite Image — Year 2026



Segmentation Map — Year 2022



Segmentation Map — Year 2026



Change Detection Map (Red=Forest Loss, Green=Forest Gain, Orange=Urbanisation)